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Perspective

Climate and society as one: a (de)coupling future for escalating climate challenges

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The escalating climate crisis demands urgent and effective action, yet the world's responses consistently fall short. Decades of research have detailed climate mechanisms, impacts, and potential technological solutions [1,2] and have informed a wide range of proposed policy responses–from carbon pricing to adaptation strategies [3]. Nevertheless, implementation has lagged far behind scientific understanding. Global emissions continue to rise; for instance, total energy-related CO_2 emissions increased by 0.8% in 2024. This trend, in turn, deepens vulnerability, particularly for marginalized communities.

A fundamental reason for this inadequacy lies in a persistent blind spot: the failure to treat climate and social systems as an integral, deeply intertwined, co-evolving entity at the core of analysis and policy-making [4,5]. The imperative to understand this climate-social coupling is one of the most critical frontiers in sustainability science. This urgency was underscored at the recent 30th Conference of the Parties (COP30) meeting in Belém, where outcomes emphasized that addressing the climate crisis requires more than biophysical solutions alone. Effective climate action must adopt a systemic approach that integrates social dimensions, including "Just Transition" frameworks. The Climate-Social System (CSS) framework presented here responds precisely to this global call for a more integrated understanding. This recognition is not without precedent. Indeed, foundational work in Social-Ecological Systems (SES), Earth System Governance (ESG), and Coupled Human-Natural Systems (CHN) has been pivotal in challenging the paradigm of a purely external human driver [6,7]. In particular, the Pan-Eurasian Experiment (PEEX) program has pioneered the development of integrated Earth System models that consider the social component as of equal importance to the natural system [8]. These frameworks provide an essential holistic worldview.

However, the core logic of these holistic perspectives has not fully permeated the dominant analytical paradigms that shape climate policy. A critical blind spot persists, not in theory, but in the conceptual architecture of our prevailing approaches. Scientific modelling and policy-making largely remain siloed, treating climatic and social processes as separate domains to be linked, rather than as components of an indivisible system. This separation is a conceptual flaw that leads to analytical shortcomings. For example, even sophisticated Integrated Assessment Models (IAMs), while linking climate and economy, are often built upon assumptions that oversimplify the complex, adaptive, and often unpredictable nature of social systems. They may capture economic reactions to climate change, but they do not inherently treat social norms, political dynamics, risk perceptions, and institutional evolution as endogenous, co-evolving variables that fundamentally shape climate trajectories [9,10].

This conceptual separation makes it difficult to adequately account for the very dynamics that research on social tipping points and adaptive governance has identified as critical [11]. The result is often a focus on incremental adjustments or technological fixes that fail to address the systemic nature of the challenge [12]. All too often, the outcome is that adaptation strategies protect infrastructure but exacerbate vulnerabilities for poorer communities [2]. For example, deforestation in Brazil's Amazon could reduce regional rainfall, threatening endemic species while simultaneously accelerating habitat loss for indigenous populations and intensifying climate warming [13]. The same study further suggests that reduced regional precipitation could result in significant economic losses in agriculture, with crop yields declining by 0.5% for each percentage point reduction in rainfall,

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thereby increasing risks of hunger and undernutrition among vulnerable populations [13]. Moreover, intensified droughts increase wildfire risk, impair carbon sequestration, and pose serious threats to human and livestock health outcomes [1]. These cascading impacts remain difficult to capture because the climate-social feedback loop that connects resource distribution to climate resilience is not treated as a central, dynamic property of the system.

To address this persistent conceptual blind spot, this perspective piece introduces, defines, and argues for the CSS as a new theoretical framework. The novelty of the CSS framework does not lie in simply acknowledging coupling, but in positing the intertwined climate-social dyad as the fundamental unit of analysis. It advocates for a conceptual reorientation: moving from studying climate and society to studying the CSS itself. This shift in perspective is what distinguishes it from prior frameworks. It compels us to ask different questions and seek explanations in the emergent, co-evolutionary properties of the system as a whole. To illustrate the analytical power that this shift in perspective affords, below we re-contextualize the classic Environmental Kuznets Curve (EKC), not as a new method, but as a demonstration of how the CSS lens can yield new insights. Ultimately, in proposing this framework, we seek to reorient research priorities toward a more systemic understanding of our coupled climate future.

The imperative of climate-social coupling. The conventional "climate risk-societal response" model is overly linear and simplistic, failing to capture the continuous and dynamic interplay through which society does not simply respond to climate change but actively shapes it, and *vice versa*. Consider the inherent complexities: public perception influences policy, policy shapes investment, investment drives technological change, and new technologies alter emissions pathways, which then feedback into the climate system, generating new impacts and responses for both people and the planet [1].

A coupling CSS is essential for governance. We define "governance alignment" as the state where institutional rules, policies, and actions (i.e., governance) are co-designed and implemented in a way that is structurally consistent with the biophysical realities and social dynamics of the CSS. However, governance itself is often actively decoupling, becoming increasingly misaligned with the CSS systems it seeks to influence. Vertically, global climate governance often struggles to translate commitments such as those embodied in the Paris Agreement into effective national implementation due to divergent interests, commitments, and capacities. Horizontally, coordination across sectors (energy, agriculture, transport, finance, health) is often weak. Strategically, short-term political cycles clash with the long-term nature of climate change resilience, creating a "collective action trap" whereby necessary transformations are delayed. Examples include internal EU disagreements over vehicle emission standards hampering the "Fit for 55" package. Italy, for instance, citing automotive manufacturing sector protection (6.2% of GDP, 2021), advocates for the postponement of the 2035 internal combustion engine (ICE) prohibition [14]. Additionally, progress on climate finance for vulnerable nations has not met stated timelines, exemplified by the delayed delivery of $115.9 billion in climate funds scheduled for 2022. These are not merely implementation failures; they are symptoms of a governance paradigm that does not internalize the deep coupling of climate and society.

Coupling the climate-social system. Climate and society do not operate as isolated domains but constitute an organic whole interconnected through material cycles, energy flows, and human activities and their consequences. Multidirectional feedback mechanisms create an inseparable two-way coupling system. For example, carbon-induced climate warming triggers agricultural productivity losses, increased food prices, and population displacement, while societal responses such as renewable energy transitions exert reciprocal influences on the climate system, forming dynamic equilibrium chains. These mechanisms are not, however, spatially confined. The CSS framework must also account for telecoupling dynamics—socioeconomic and environmental interactions over distances—whereby consumption, trade, and policies in one region drive profound environmental and social outcomes (e.g., land-use change or emissions) in another [15]. Integrating telecoupling provides a robust theoretical foundation for analyzing the cross-scale feedback loops and distal interactions that are fundamental to our CSS concept. Understanding this coupling system hinges on three core characteristics: nonlinear threshold effects that govern system evolution, spatiotemporal scale nestedness demanding multi-tiered responses, and convergent innovation in knowledge production that enables effective action.

Research priorities for CSS should therefore focus on three key areas. First, developing coupling models that integrate not only natural processes but also societal decision-making is crucial to identify intervention leverage points at critical nodes. Second, creating dynamic risk assessment frameworks is needed to quantify both the societal costs of climate policies and the emission-reduction potential of adaptive measures. Third, constructing open-access data platforms is essential to enable the deep integration of meteorological observations, modeled predictions, economic statistics, and human outcome and behavioral big data. Only through a systemic understanding of this symbiotic relationship can we design sustainable development pathways that synergize climate benefits with societal viability.

Strong, weak and de-coupling. The EKC established the foundation for understanding climatic, environmental and socio-economic relationships. However, its implicit assumption of "technological determinism" *viz.* that economic growth inevitably triggers an emissions reduction turning point, fails to explain why comparable economies often exhibit divergent trajectories (e.g., Canada and Qatar, both high-income countries but with opposing carbon emission trends) and whether EKC has the possibility of regression.

Utilizing two core indicators, log-transformed GDP per capita and log-transformed CO_2 emissions, we characterize each economy's carbon emission–economic growth trajectory (Fig. 1a–d) using quadratic polynomial curve fitting. We then analyze these trajectories in the context of their Nationally Determined Contributions (NDCs). By "re-contextualizing" the EKC, we transform the classical curve into a dynamic coupling illustrative framework for climate-social systems.

Fig. 1 illustrates dynamic climate-social coupling states and their relationship to stated climate ambitions. Panel (a) provides the historical baseline (1899–1959). During this early industrial period, developed nations (e.g., the United States, United Kingdom, Germany) were uniformly on a steep ascent, characterizing a "weak coupling" state where economic industrialization was tightly linked to rising emissions. Panel (b) shows the modern era (1960–2023) for these same nations, illustrating their transition to "strong coupling". These nations (e.g., United States t_f: 1981; Germany t_f: 1974) have passed their emission peaks (t_f) and now demonstrate a benign coupling of sustained growth with declining emissions. Their ambitious 2050 carbon neutrality targets are thus aligned with their demonstrated post-peak trajectories. Panel (c) presents the complex modern trajectories for developing and emerging economies. Here, the (mis)alignment between trajectory and ambition becomes crucial. Economies like China and India are on "weak coupling" ascent phases, similar in shape to the historical trajectories in Panel (a). Their 2060–2070 neutrality targets reflect this pre-peak status, acknowledging the developmental phase of their coupling. Nations like Qatar, Iraq, and Kenya display structurally divergent paths-either "decoupling" (anti-EKC) or low-emission trajectories. Their 2030-focused goals

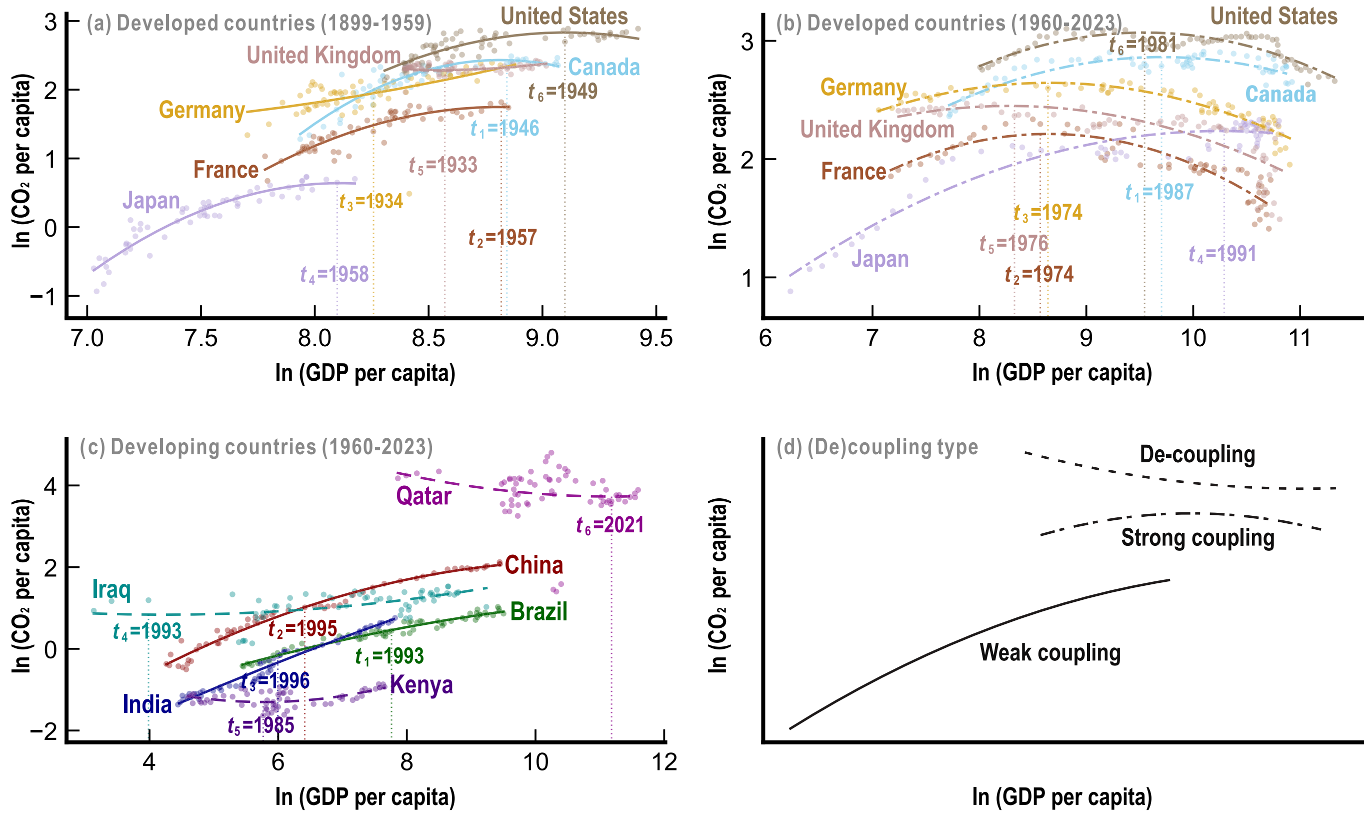

Fig. 1. Climate-social (de)coupling states across diverse economies and historical periods. (a) Historical trajectories (1899–1959) for selected developed economies, showing a pre-peak "weak coupling" state. (b) Modern trajectories (1960–2023) for the same developed economies, illustrating the transition to a "strong coupling" state. (c) Modern trajectories (1960–2023) for selected developing and emerging economies. (d) Conceptual synthesis of the three (de)coupling types observed in panels a, b, and c, where "strong coupling" typically represents post-peak economies with sustained growth and declining emissions, "weak coupling" denotes pre-peak economies with continued growth and slowing emission increases, and "de-coupling" corresponds to fossil-fuel-dependent, low-income economies requiring structural transformation. NDCs Targets: The trajectories are analyzed against stated climate goals: 2050 Carbon Neutrality: US, UK, Canada, Germany, France, Japan (Panels a and b), and Brazil (Panel c). 2060–2070 Neutrality: China, India (Panel c). 2030/2035 Reduction Goals (Qatar: −25% vs. 2019; Kenya: −32% vs. Business as Usual (BAU); Iraq: −15% vs. BAU): Qatar, Iraq, Kenya (Panel c). Note: All panels plot the natural logarithm of CO_2 emissions per capita (ln (CO_2 per capita)) against the natural logarithm of GDP per capita (ln (GDP per capita)), fitted with quadratic polynomial regressions. GDP data are sourced from the Maddison Database (1899–1959) and the World Bank (1960–2023), while CO_2 data are from the Global Carbon Budget. To preserve internal consistency given methodological differences, historical and modern series are charted separately; the log transformation ensures that within-series trend analyses are unaffected by these cross-source differences. Additionally, to prevent potential ambiguity, we clarify our specific definition of "decoupling". In contrast to its conventional positive connotation in climate economics, our framework re-contextualizes the term to describe a negative state: an "anti-EKC" trajectory indicating a structural failure to align economic development with climate resilience.

(lacking long-term neutrality pledges) reflect deep-seated constraints, such as fossil-fuel dependency or structural poverty, that inhibit systemic alignment. Most notably, Brazil's case highlights the tension within the CSS. Although Brazil shares an ambitious 2050 neutrality target with the "strong coupling" nations in Panel (b), its actual emissions trajectory (a "weak coupling" ascent) places it firmly in Panel (c) alongside China and India. This misalignment between an ambitious goal and a challenging developmental trajectory underscores the significant structural transformation required for such a goal to be realized, demonstrating a potential "decoupling" between governance ambition and systemic reality. Finally, Panel (d) synthesizes these empirical patterns observed in (a), (b), and (c) into our conceptual framework.

The three climate-society coupling states—strong, weak, and decoupled—are not static or linear but dynamic and reversible. Indeed, governance serves as the critical steering mechanism that determines the direction and speed of these transitions. For instance, Trump's "drill, baby, drill" slogan and his 2017 and 2025 withdrawals from the Paris Agreement [16] risk weakening U.S. climate-social coupling by prioritizing fossil fuel expansion over systemic resilience, thereby escalating climate vulnerabilities and increasing societal exposure to climate-related risks. Such shifts demonstrate how policy reversals can degrade institutional coherence, transforming strong coupling into weak or even decoupled states. Conversely, positive trajectories are equally evident. Emerging economies, notably China, exemplify the potential to accelerate the transition from weak toward strong coupling. This progress is underpinned by robust policy commitments, such as extensive purchase subsidies and tax exemptions for New Energy Vehicles (NEVs), aligned with its "Dual Carbon" targets. The impact of this policy-driven technological adoption is profound: by 2024, China's NEV fleet grew to 31.4 million (accounting for 8.9% of the total car park) [17]. More significantly, NEVs constituted 41.83% of all new vehicle registrations, indicating a rapid systemic shift in transport. This demonstrates how targeted policy and rapid technological uptake can drive a nation to actively progress toward a "strong coupling" state, bending its emissions curve.

A (de)coupling future. Addressing the escalating challenge of climate change calls for an approach that acknowledges the profound coupling between climate processes and social systems. Traditional research paradigms and governance models that treat these systems in isolation have proven increasingly inadequate for confronting the accelerating crisis. Future research grounded in the CSS framework should prioritize identifying coupling mechanisms through quantitative studies and scenario-based analyses of climate–social feedback across scales (from global to local) and sectors (including energy, transport, agriculture and food systems, finance, health, and culture). Achieving this, however, demands more than a simple "plug-and-play" process of linking disparate models. It necessitates a fundamental shift toward "team science" and genuine knowledge co-production [18]. This implies moving to iterative, participatory frameworks where interdisciplinary teams—spanning geography (bridging human-environment systems), climate science (providing biophysical grounding), public administration (informing governance pathways), political science, economics, and sociology—co-design the model architecture, core assumptions, and feedback mechanisms from the very outset. This co-production approach is essential to bridge the deep epistemological and methodological divides that often separate biophysical process modeling from the analysis of governance feasibility and human behavior. For instance, this could involve moving beyond traditional IAMs by coupling them with agent-based models (ABMs) to better simulate the complex social feedback and human behaviors we identify. Furthermore, the CSS concept could be operationalized within emerging international frameworks, such as the Earth Commission's work on "Safe and Just Earth System Boundaries", which explicitly seeks to integrate biophysical limits with social justice indicators.

A particular focus should be placed on detecting tipping points, leverage points, and potential pathways toward systemic collapse or transformation. The development of innovative governance tools and practices is also important with CSS-informed adaptive, polycentric, and participatory governance models. Concrete examples include the implementation of national-level "Citizens" Assemblies on Climate (as seen in France and the UK) or operationalizing the CSS framework through international mechanisms like the "Just Transition" funds and dialogues. This work must, in turn, be supported by next-generation open data platforms, such as the "OS-Climate" (Open Source Climate) initiative, which is building the open data infrastructure required to integrate the very biophysical, economic, and social data our framework demands.

In conclusion, addressing future climate challenges requires an approach that fully acknowledges and engages with the intrinsic coupling between climate and social systems. In an era marked by trade disputes, geopolitical instability, and transformative technologies such as artificial intelligence, the dynamics—and future—of this relationship face unprecedented uncertainty. Understanding the range of coupling states, from strong integration to weak linkage or outright decoupling, at national and regional levels is critically important**.** Achieving and sustaining strong climate-social coupling is therefore central to building the societal resilience required to effectively confront mounting climate change pressures.

Conflict of interest

The authors declare that they have no conflict of interest.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scib.2026.03.026.

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